Panel d cluster analysis summary
N = 50,000 timesteps (phase-1 random walk)
PCA variance: PC1=31.8%, PC2=11.8%
Silhouette k=3 (KMeans on PCA): 0.845

Partition agreement (ARI):
  KMeans vs x-tertile: -0.000
  x-tertile vs wall proximity: 0.142
  KMeans vs PC1-tertile: 1.000

Visual lobes (x-tertile) — top drivers:
  x: η²=0.875
  spatial_bin: η²=0.875
  local_density: η²=0.355
  dist_reward: η²=0.273
  dist_center: η²=0.273

KMeans (PC1 bands) — top drivers:
  obs_13: η²=0.0001
  obs_15: η²=0.0001
  obs_03: η²=0.0001

R² PC2 from [x,y]: 0.001 from [x,y,θ,obs]: 0.006